

Task 1: To Detect Background Jobs, Launch Non-Blocking Processes, Return Prompt Immediately, Store Job Information, Monitor Execution, Test Background Tasks

```
GNU nano 7.2 task1.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        printf("Background job started. PID = %d\n", getpid());

        sleep(10);

        printf("Background job completed. PID = %d\n", getpid());

        exit(0);
    }

    printf("Job launched in background.\n");
}
```

```
cd ~/background_jobs
shreyas@Grigori:~/background_jobs$ nano task1.c
shreyas@Grigori:~/background_jobs$ gcc task1.c -o task1
shreyas@Grigori:~/background_jobs$ ./task1 &
[1] 913
shreyas@Grigori:~/background_jobs$ Job launched in background.
Job PID: 914
Parent continues immediately.
Background job started. PID = 914
^C
[1]+  Done                  ./task1
shreyas@Grigori:~/background_jobs$ jobs
Background job completed. PID = 914
shreyas@Grigori:~/background_jobs$ jobs
shreyas@Grigori:~/background_jobs$ ps
  PID TTY          TIME CMD
  883 pts/0    00:00:00 bash
  915 pts/0    00:00:00 ps
shreyas@Grigori:~/background_jobs$ ps -p 914
  PID TTY          TIME CMD
shreyas@Grigori:~/background_jobs$ sleep 20 &
[1] 917
shreyas@Grigori:~/background_jobs$ jobs
[1]+  Running                  sleep 20 &
shreyas@Grigori:~/background_jobs$ kill %1
-bash: kill: (917) - No such process
[1]+  Done                    sleep 20
shreyas@Grigori:~/background_jobs$ jobs
shreyas@Grigori:~/background_jobs$ rm -f task1
shreyas@Grigori:~/background_jobs$ |
```

Task 2: To Create Job Table, Store Process Groups, Track States, Update Status, Remove Completed Jobs, Validate Job Records

```
GNU nano 7.2 task2.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_JOBS 10

typedef struct {
    int job_id;
    pid_t pid;
    pid_t pgid;
    int active;
} Job;

Job jobs[MAX_JOBS];
int job_count = 0;

void add_job(pid_t pid) {
    if (job_count >= MAX_JOBS) {
        printf("Job table full.\n");
        return;
    }

    jobs[job_count].job_id = job_count + 1;
    jobs[job_count].pid = pid;
    jobs[job_count].pgid = getpgid(pid);
```

```
    printf("Job added: ID=%d PID=%d PGID=%d\n",
           jobs[job_count].job_id,
           jobs[job_count].pid,
           jobs[job_count].pgid);

    job_count++;
}

void update_jobs() {
    for (int i = 0; i < job_count; i++) {
        if (!jobs[i].active)
            continue;

        int status;

        pid_t result = waitpid(
            jobs[i].pid,
            &status,
            WNOHANG
        );

        if (result == 0) {
            printf("Job %d still running.\n",
                   jobs[i].job_id);
        }
    }
}
```

```

void show_jobs() {
    printf("\nJob Table:\n");

    for (int i = 0; i < job_count; i++) {
        printf(
            "Job ID: %d | PID: %d | PGID: %d | State: %s\n",
            jobs[i].job_id,
            jobs[i].pid,
            jobs[i].pgid,
            jobs[i].active ? "RUNNING" : "COMPLETED"
        );
    }
}

int main() {

    for (int i = 0; i < 3; i++) {

        pid_t pid = fork();

        if (pid < 0) {
            perror("fork");
            return 1;
        }
    }
}

```

```

shreyas@Grigori:~/background_jobs$ nano task2.c
shreyas@Grigori:~/background_jobs$ gcc task2.c -o task2
shreyas@Grigori:~/background_jobs$ ./task
-bash: ./task: No such file or directory
shreyas@Grigori:~/background_jobs$ ./task2
Job added: ID=1 PID=928 PGID=927
Child process started: PID=928
Job added: ID=2 PID=929 PGID=927
Child process started: PID=929
Job added: ID=3 PID=930 PGID=927

Job Table:
Job ID: 1 | PID: 928 | PGID: 927 | State: RUNNING
Job ID: 2 | PID: 929 | PGID: 927 | State: RUNNING
Job ID: 3 | PID: 930 | PGID: 927 | State: RUNNING
Child process started: PID=930

Checking job status...
Job 1 still running.
Job 2 still running.
Job 3 still running.

Job Table:
Job ID: 1 | PID: 928 | PGID: 927 | State: RUNNING
Job ID: 2 | PID: 929 | PGID: 927 | State: RUNNING
Job ID: 3 | PID: 930 | PGID: 927 | State: RUNNING
Child process completed: PID=928

Checking job status...
Job 1 completed and removed.

```

```

Job ID: 3 | PID: 930 | PGID: 927 | State: RUNNING
Child process completed: PID=929
Child process completed: PID=930

Checking job status...
Job 2 completed and removed.
Job 3 completed and removed.

Job Table:
Job ID: 1 | PID: 928 | PGID: 927 | State: COMPLETED
Job ID: 2 | PID: 929 | PGID: 927 | State: COMPLETED
Job ID: 3 | PID: 930 | PGID: 927 | State: COMPLETED

Checking job status...

Job Table:
Job ID: 1 | PID: 928 | PGID: 927 | State: COMPLETED
Job ID: 2 | PID: 929 | PGID: 927 | State: COMPLETED
Job ID: 3 | PID: 930 | PGID: 927 | State: COMPLETED

All jobs completed.

Job Table:
Job ID: 1 | PID: 928 | PGID: 927 | State: COMPLETED
Job ID: 2 | PID: 929 | PGID: 927 | State: COMPLETED
Job ID: 3 | PID: 930 | PGID: 927 | State: COMPLETED
shreyas@Grigori:~/background_jobs$ ps -ef | grep task2
shreyas      958      883  0 14:30 pts/0    00:00:00 grep --color=auto task2
shreyas@Grigori:~/background_jobs$ rm -f task2
shreyas@Grigori:~/background_jobs$ |

```

```

shreyas@Grigori: ~
shreyas@Grigori:~$ ps -ef | grep task2
shreyas      955      937  0 14:29 pts/2    00:00:00 grep --color=auto task2
shreyas@Grigori:~$ ps -o pid,ppid,pgid,state,cmd
  PID   PPID   PGID S  CMD
   937    932    937 S  -bash
   956    937    956 R  ps -o pid,ppid,pgid,state,cmd
shreyas@Grigori:~$ |

```